Aalborg University
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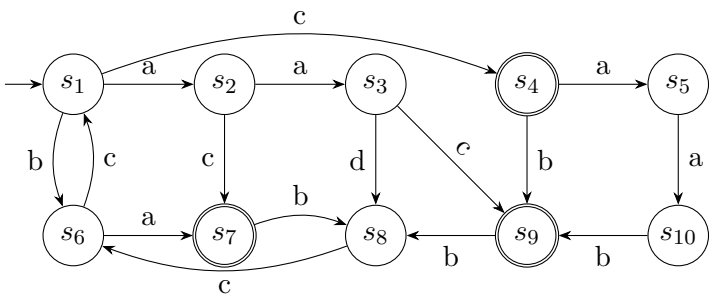
Machine Intelligence
Fall 2025

Exercise Sheet 4

1 Abstractions

1.1. Consider the state space Θ depicted below.



Let the cost function function be as follows:

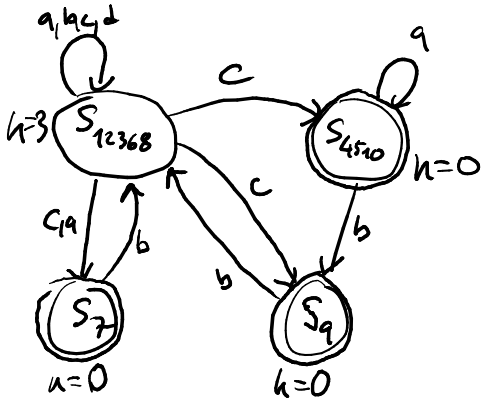
action	cost
a	3
b	1
c	4
d	0

Consider two abstraction functions α_1 and α_2 defined as follows:

state s	$\alpha_1(s)$	$\alpha_2(s)$
s_1	1	x
s_2	1	y
s_3	1	z
s_4	2	w
s_5	2	w
s_6	1	x
s_7	3	y
s_8	1	z
s_9	4	w
s_{10}	2	w

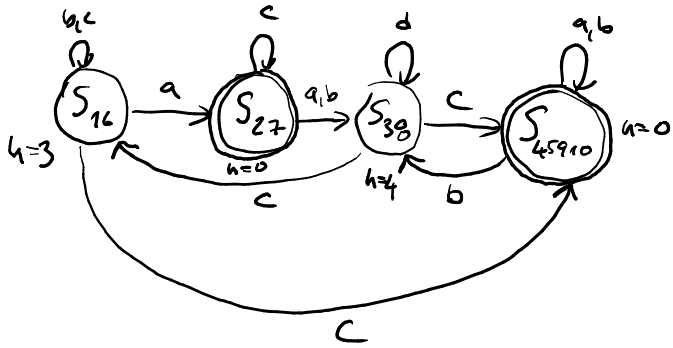
(a) Draw the abstract state space Θ^{α_1} and annotate each abstract state with its h^*-value in Θ^{α_1}.

Solution: Abstract states annotated with concrete states' indexes, i.e., $s_{12368} = 1, s_{4510} = 2, s_7 = 3, s_9 = 4$.



(b) Draw the abstract state space Θ^{α_2} and annotate each abstract state with its h^*-value in Θ^{α_2}.

Solution:



(c) Fill the following table with the correct values of $h^{\alpha_1}(s)$ and $h^{\alpha_2}(s)$ for all states s in Θ:

state s	$h^{\alpha_1}(s)$	$h^{\alpha_2}(s)$
s_1		
s_2		
s_3		
s_4		
s_5		
s_6		
s_7		
s_8		
s_9		
s_{10}		

Solution:

state s	$h^{\alpha_1}(s)$	$h^{\alpha_2}(s)$
s_1	3	3
s_2	3	0
s_3	3	4
s_4	0	0
s_5	0	0
s_6	3	3
s_7	0	0
s_8	3	4
s_9	0	0
s_{10}	0	0

2 PDBs

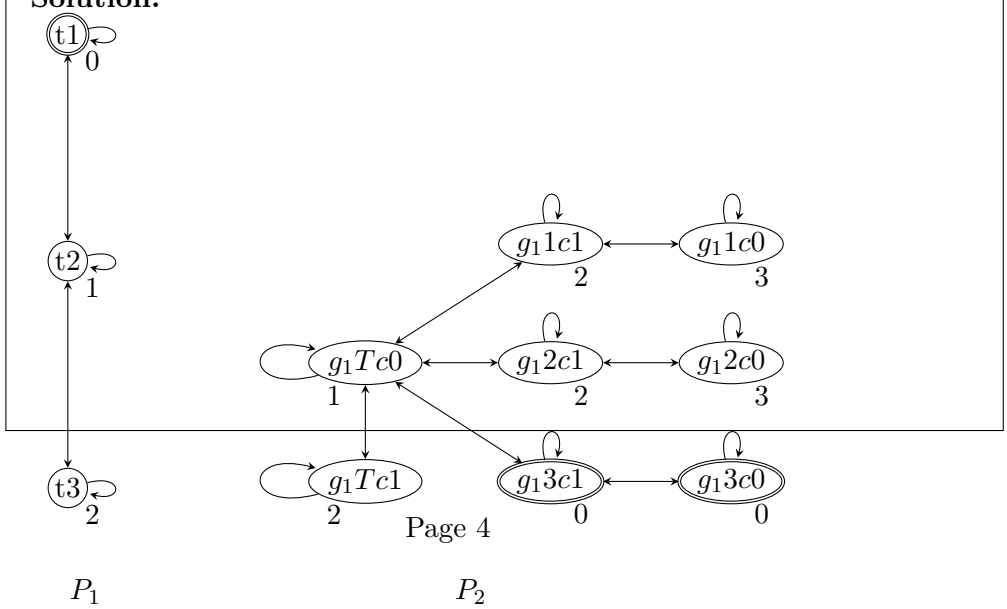
2.1. Consider the following **Xmax Transportation** problem where a Xmas truck has to transport two gifts to their goal location. At any point in time only one package can be loaded into the truck. Formally, the task $\Pi = (\mathcal{V}, \mathcal{A}, \text{cost}, I, G)$ is encoded in FDR as follows:

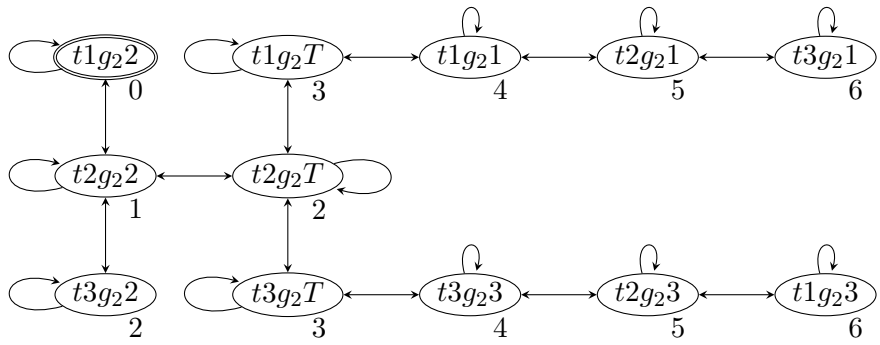
- $\mathcal{V} = \{t, g_1, g_2, c\}$ with domains
  - $D_t = \{1, 2, 3\}$,
  - $D_{g_1} = D_{g_2} = \{1, 2, 3, T\}$,
  - $D_c = \{0, 1\}$ (the capacity of the truck)
- $\mathcal{A} = \{drive(x, y), load(i, z), unload(i, z)\}$ (all actions have unit cost) for $\{x, y\} \in \{\{1, 2\}, \{2, 3\}\}$, $i \in \{1, 2\}$, and $z \in \{1, 2, 3\}$
  - $\text{pre}_{drive(x,y)} = \{t = x\}$,
    $\text{eff}_{drive(x,y)} = \{t = y\}$
  - $\text{pre}_{load(i,z)} = \{t = z, g_i = z, c = 1\}$,
    $\text{eff}_{load(i,z)} = \{g_i = T, c = 0\}$
  - $\text{pre}_{unload(i,z)} = \{t = z, g_i = T, c = 0\}$,
    $\text{eff}_{unload(i,z)} = \{g_i = z, c = 1\}$
- $I = \{t = 1, g_1 = 2, g_2 = 3, c = 1\}$,
- $G = \{t = 1, g_1 = 3, g_2 = 2\}$.

Consider the patterns $P_1 = \{t\}$, $P_2 = \{g_1, c\}$, and $P_3 = \{t, g_2\}$.

(a) Compute a pattern database for each of the three patterns. To do so, execute the algorithm given in the lecture: Construct the reachable state space $\Theta_{\Pi|_{P_i}}$ of the syntactic projection onto P_i by a breadth-first forward search. Give the states, abbreviating their variable values (e.g., the initial state of $\Theta_{\Pi|_{P_i}}$ can be notated $t1g_23$), and draw an edge from each state to each of its successor states. Annotate each of the states in $\Theta_{\Pi|_{P_i}}$ with its goal distance.

Solution:





P_3

(b) Build the compatibility graph for $\mathcal{C} = \{P_1, P_2, P_3\}$. What is the value of $h^{\mathcal{C}}(I)$? Explain how you got to the values.

Solution:

P_1 ———————— P_2 $\qquad\qquad$ P_3

Only P_1 and P_2 are orthogonal. Thus, $h^{\mathcal{C}}(I) = \max\{h^{P_1}(I) + h^{P_2}(I), h^{P_3}(I)\} = \max\{0 + 2, 6\} = 6$.

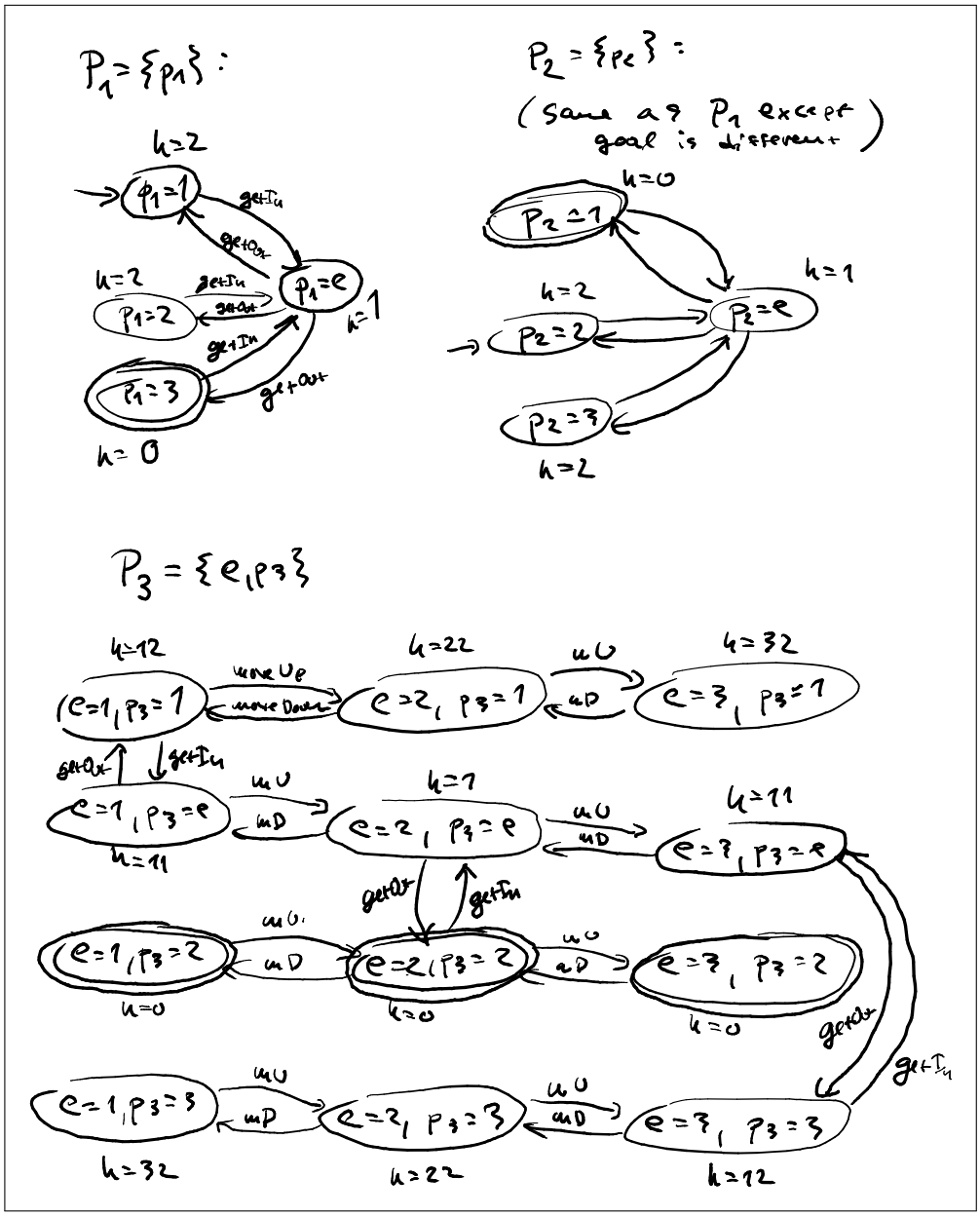
2.2. Consider the following **Elevator** FDR planning task.

- $\mathcal{V} = \{e, l, p_1, p_2, p_3\}$, $D_e = \{1, 2, 3\}$, $D_l = \{0, 1, 2\}$, $D_{p_i} = \{1, 2, 3, e\}$ for $i \in \{1, 2, 3\}$
  - e: current floor of the elevator
  - l: number of people in the elevator
  - p_i: location of person i (on the floor or inside the elevator)
- $I = \{e{=}1, l{=}0, p_1{=}1, p_2{=}2, p_3{=}3\}$
- $G = \{p_1{=}3, p_2{=}1, p_3{=}2\}$
- $\mathcal{A} = \{moveUp(x), moveDown(x), getIn(x, y, z), getOut(x, y, z)\}$ where
  - $moveUp(x)$ for $x \in \{1, 2\}$
    - pre: $e{=}x$
    - eff: $e{=}x + 1$
    - cost: 10
  - $moveDown(x)$ for $x \in \{2, 3\}$
    - pre: $e{=}x$
    - eff: $e{=}x - 1$
    - cost: 10
  - $getIn(x, y, z)$ for $x \in \{1, 2, 3\}$, $y \in \{1, 2, 3\}$, $z \in \{0, 1\}$
    - pre: $e{=}x, p_y{=}x, l{=}z$
    - eff: $p_y{=}e, l{=}z + 1$
    - cost: 1
  - $getOut(x, y, z)$ for $x \in \{1, 2, 3\}$, $y \in \{1, 2, 3\}$, $z \in \{1, 2\}$
    - pre: $e{=}x, p_y{=}e, l{=}z$
    - eff: $p_y{=}x, l{=}z - 1$
    - cost: 1

Consider the patterns $P_1 = \{p_1\}$, $P_2 = \{p_2\}$, and $P_3 = \{e, p_3\}$.

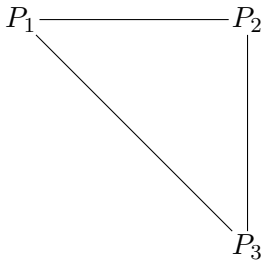
(a) Compute a pattern database for each of the three patterns. To do so, execute the algorithm given in the lecture: Construct the reachable state space $\Theta_{\Pi|_{P_i}}$ of the syntactic projection onto P_i by a breadth-first forward search. Give the states, abbreviating their variable values, and draw an edge from each state to each of its successor states. Annotate each of the states in $\Theta_{\Pi|_{P_i}}$ with its goal distance.

Solution:



(b) Build the compatibility graph for $\mathcal{C} = \{P_1, P_2, P_3\}$. What is the value of $h^{\mathcal{C}}(I)$? Explain how you got to the values.

Solution:



$h^{P_1}(I) = 2$
$h^{P_2}(I) = 2$
$h^{P_3}(I) = 32$
$h^{\mathcal{C}}(I) = h^{P_1}(I) + h^{P_2}(I) + h^{P_3}(I) = 36$